

ECx00G&EC600U&EG800G Series

QuecOpen(SDK) Decoding Development Guide

LTE Standard Module Series

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About the Document

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1 Introduction

Quectel EC600U series, EG800G series and ECx00G family modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS, which is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]**.

This document is applicable to QuecOpen® solution based on SDK build environment. It introduces decoding APIs, the relevant example and debugging of EC600U series, EG800G series and ECx00G family modules in QuecOpen® solution.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC600U Series
ECx00G	EC200G-CN
	EC600G Series
	EC800G-CN
-	EG800G Series

2 Decoding APIs

2.1. Header File

`ql_api_decoder.h`, the header file of decoding APIs, is in the `components\ql-kernel\inc` directory.

2.2. API Overview

Table 2: API Overview

Function	Description
<code>ql_qr_decoder_init()</code>	Initializes decoding.
<code>ql_qr_image_decoder()</code>	Decodes the inputted barcode or QR code.
<code>ql_qr_get_decoder_result()</code>	Gets the decoding result.
<code>ql_destroy_decoder()</code>	Releases decoding.
<code>ql_get_decoder_version()</code>	Gets decoding library version.

2.3. API Description

2.3.1. `ql_qr_decoder_init`

This function initializes decoding.

- **Prototype**

```
ql_errcode_decoder_e ql_qr_decoder_init(void)
```

- **Parameter**

None

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.1.1. ql_errcode_decoder_e

The enumeration of execution results:

```
typedef enum
{
    QL_DECODER_SUCCESS = QL_SUCCESS,
    QL_DECODER_INIT_ERR           = 1|QL_COMPONENT_BSP_DECODER,
    QL_DECODER_ERR,
    QL_DECODER_GET_RESULT_ERR,
    QL_DECODER_GET_RESULT_LENGTH_ERR,
    QL_DECODER_DESTROY_ERR,
}ql_errcode_decoder_e;
```

- **Member**

Member	Description
QL_DECODER_SUCCESS	Successful execution.
QL_DECODER_INIT_ERR	Failed to initialize decoding.
QL_DECODER_ERR	Failed to decode.
QL_DECODER_GET_RESULT_ERR	Failed to get the decoding results.
QL_DECODER_GET_RESULT_LENGTH_ERR	Failed to get the length of decoding results.
QL_DECODER_DESTROY_ERR	Failed to release decoding.

2.3.2. ql_qr_image_decoder

This function decodes the inputted barcode or QR code.

- **Prototype**

```
ql_errcode_decoder_e ql_qr_image_decoder(uint16_t *img_buffer, uint32_t width, uint32_t height)
```

- **Parameter**

**img_buffer:*

[In] Inputted images.

width:

[In] Width of inputted image.

height:

[In] Height of inputted image.

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.3. ql_qr_get_decoder_result

This function gets the decoding result.

- **Prototype**

```
ql_errcode_decoder_e ql_qr_get_decoder_result (ql_decoder_type_e* type, unsigned char * result);
```

- **Parameter**

**type:*

[Out] Type of decoding. See **Chapter 2.3.3.1** for details.

**result:*

[Out] Buffer for storing the result of scanning.

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.3.1. ql_decoder_type_e

The enumeration of decoding types:

```
typedef enum
{
    QL_DECODER_TYPE_CODE39 = 0,
    QL_DECODER_TYPE_CODE128,
    QL_DECODER_TYPE_QR_CODE,
    QL_DECODER_TYPE_NONE = 0xff,
```

```
}ql_decoder_type_e;
```

- **Member**

Member	Description
<code>QL_DECODER_TYPE_CODE39</code>	Code-39.
<code>QL_DECODER_TYPE_CODE128</code>	Code-128.
<code>QL_DECODER_TYPE_QR_CODE</code>	QR code.
<code>QL_DECODER_TYPE_NONE</code>	No recognition.

2.3.4. ql_destroy_decoder

This function releases decoding and releases the memory space requested by the decoder.

- **Prototype**

```
ql_errcode_decoder_e ql_destroy_decoder(void)
```

- **Parameter**

None

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.5. ql_get_decoder_version

This function gets the current version of the decoding library.

- **Prototype**

```
ql_errcode_decoder_e ql_get_decoder_version(unsigned char* version);
```

- **Parameter**

**version:*

[Out] The current version of decoding library.

- **Return Value**

See **Chapter 2.3.1.1** for details.

3 Development Example and Debugging

3.1. Development Example of App Decoding

3.1.1. Static Library Use for Decoding

The module decoding library is used as a static library, therefore you need to link the static library when using it. The static library address is stored in *components/newlib/armca5/libq_l_api_decoder.a*. You only need to link the static library where the library is required. Please see *CmakeLists.txt* in the root directory of App SDK for details, as shown in the following figure:



```

if(CONFIG_APPIMG_LOAD_FLASH)
    set(target ${QL_APP_BUILD_VER})
    add_appimg_flash_q_l_example(${target} q_l_init.c)
    target_link_libraries(${target} PRIVATE q_l_app_nw q_l_app_peripheral q_l_app_osi q_l_app_dev q_l_app_sim q_l_app_power)
endif()
if(QL_APP_FEATURE_DECODER)
    add_library(q_l_decoder_api STATIC IMPORTED)
    set_target_properties(q_l_decoder_api PROPERTIES IMPORTED_LOCATION ${SOURCE_TOP_DIR}/components/newlib/armca5/libq_l_api_decoder.a)
    target_link_libraries(${target} PRIVATE q_l_app_decoder q_l_decoder_api ${libm_file_name})
endif()
if(QL_APP_FEATURE_FTP)
    target_link_libraries(${target} PRIVATE q_l_app_ftp)
endif()
if(QL_APP_FEATURE_HTTP)
    target_link_libraries(${target} PRIVATE q_l_app_http)
endif()
    
```

The App using decoding library Link static library

Figure 1: Static Library Linking Example

3.1.2. Decoding Example

decoder_demo.c, the decoding example file, is provided in the SDK of Quectel EC600U series, EG800G series and ECx00G family modules, in the *components/q_l-application/decoder* directory.

The example mainly includes:

- Initializing decoding
- Scanning codes with camera
- Decoding
- Getting decoding results

The example entry function is *q_l_decoder_app_init()*, as shown below:

```

83: void ql_decoder_app_init(void)
84: {
85:     QIDECODERStatus err = QL_OSI_SUCCESS;
86:     ql_task_t decoder_task = NULL;
87:
88:     err = ql_rtos_task_create(&decoder_task, QL_DECODER_TASK_STACK_SIZE, QL_DECODER_TASK_Prio, "decoder DEMO", ql_decoder_demo_thread, NULL, QL_DECODER_TASK_EVENT_CNT);
89:     if (err != QL_OSI_SUCCESS)
90:     {
91:         QL_DECODER_LOG("DECODER demo task created failed");
92:     }
93: }
94:

```

Figure 2: Entry Function of Decoding Example

In the example, decoding is initialized first, then the camera and LCD are turned on with preview enabled, and then code scanning, decoding and getting results are started in sequence.

3.1.3. Example Auto-Start

The above example is not started by default in *ql_init_demo_thread()*, as shown below. If necessary, uncomment *ql_decoder_app_init()* to start the example.

```

static void ql_init_demo_thread(void *param)
{
    QL_INIT_LOG("init demo thread enter, param 0x%x", param);
    #if 0
        ql_gpio_app_init();
        ql_gpioint_app_init();
    #endif

    #ifdef QL_APP_FEATURE_DECODER
        //ql_decoder_app_init();
    #endif

    #ifdef QL_APP_FEATURE_LEDCFG
        ql_ledcfg_app_init();
    #endif

    #ifdef QL_APP_FEATURE_AUDIO
        //ql_audio_app_init();
    #endif

    #ifdef QL_APP_FEATURE_LCD
        //ql_lcd_app_init();
    #endif
    #ifdef QL_APP_FEATURE_LVGL
        //ql_lvgl_app_init();
    #endif
}

```

Figure 3: Setting of Decoding Example Auto-Start

NOTE

The camera is required in decoding, so you need to uncomment *ql_ledcfg_app_init()* when you use the example. And some functionality pins required by the camera are also used in *ql_ledcfg_app_init()*, resulting in conflicts and abnormal camera operation. In addition, when using this example, do not execute *ql_camera_app_init()* or *ql_i2c_demo_init()* at the same time, because all three examples use cameras for demonstration purposes. If all three entry functions are started at the same time, they will conflict with each other and cause module exceptions.

3.2. Debugging of Decoding

3.2.1. Debugging Procedure

The decoding can be debugged on Quectel LTE OPEN EVB. The debugging procedure is as follows:

Download the compiled firmware to the module. Connect the USB port of the LTE OPEN EVB to a PC with a USB cable. USB AP Log Port and USB DIAG Port mainly displays the system debugging information. You can view the information of the example through USB AP Log Port or USB DIAG Port after capturing the log with cooltools or ArmLogel tool. For details about log capture, see **documents [2]** and **[3]**.

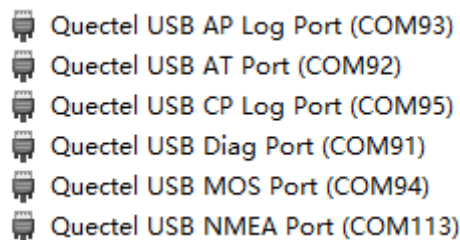


Figure 4: Ports in Device Manager 1 (EC600U Series)

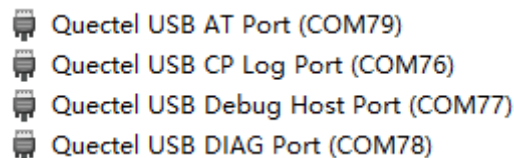


Figure 5: Ports in Device Manager 2 (EG800G Series/ECx00G Family)

NOTE

For EC600U series module, cooltools tool is used to capture logs through USB AP Log Port, see **document [2]** for details on log capture. For EG800G series and ECx00G family modules, ArmLogel tool is used to capture logs through USB DIAG Port, see **document [3]** for details on log capture.

3.2.2. Debugging Log Information

Uncomment `ql_decoder_app_init()` so that the decoding example would start automatically at bootup and judge if the decoding runs successfully in the log information. If the decoding is successful, the QR code or barcode can be recognized. Otherwise, the QR code or barcode cannot be recognized.

The debugging information on AP log port is shown in **Figure 6** and **Figure 7**.

[ql_api_qr_code][ql_qr_decoder_init, 131] Qr code Initial success
[ql_api_qr_code][ql_qr_image_decoder, 161] decoder into preview data
[ql_api_qr_code][ql_qr_image_decoder, 179] *** Decoding end***
[ql_api_qr_code][ql_qr_image_decoder, 182] Imgbuffer Decoder failed
[ql_DECODERDEMO][ql_decoder_demo_thread, 98] QR code decode failed
[ql_api_qr_code][ql_qr_image_decoder, 161] decoder into preview data
[ql_api_qr_code][ql_qr_image_decoder, 179] *** Decoding end***
[ql_api_qr_code][ql_qr_image_decoder, 182] Imgbuffer Decoder failed
[ql_DECODERDEMO][ql_decoder_demo_thread, 98] QR code decode failed

Figure 6: Log Indicating Failure to Recognize QR Code or Barcode

[ql_api_qr_code][ql_qr_image_decoder, 161] decoder into preview data
[ql_api_qr_code][ql_qr_image_decoder, 179] *** Decoding end***
[ql_api_qr_code][ql_qr_image_decoder, 206] imgbuffer Decoder success
[ql_api_qr_code][ql_qr_get_decoder_result, 247] Get Result Length success
[ql_DECODERDEMO][ql_decoder_demo_thread, 90] Get QR code decode result **34561561513131651351351303531351351351313**
[ql_api_qr_code][ql_qr_image_decoder, 161] decoder into preview data
[ql_api_qr_code][ql_qr_image_decoder, 179] *** Decoding end***
[ql_api_qr_code][ql_qr_image_decoder, 206] imgbuffer Decoder success
[ql_api_qr_code][ql_qr_get_decoder_result, 247] Get Result Length success
[ql_DECODERDEMO][ql_decoder_demo_thread, 90] Get QR code decode result **34561561513131651351351303531351351351313**

Figure 7: Log Indicating QR Code or Barcode is Recognized Successfully

4 Appendix References

Table 3: Related Documents

Document Name
[1] Quectel_ECx00G&EC600U&EG800G_Series_QuecOpen_CSDK_Quick_Start_Guide
[2] Quectel_EC600U_Series_QuecOpen_Log_Capture_Guide
[3] Quectel_ECx00G&EG800G_Series_QuecOpen_Log_Capture_Guide

Table 4: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
APP	Application
COM	Communication
EVB	Evaluation Board
IoT	Internet of Things
LCD	Liquid Crystal Display
LTE	Long-Term Evolution
PC	Personal Computer
RTOS	Real-Time Operating System
SDK	Software Development Kit
USB	Universal Serial Bus